

Experiments / 实验

BNPDFormer Main Results, Ablation Studies, and Baseline Comparisons

BNPDFormer 主模型结果、消融实验与基线对比

The experiments report BNPDFormer on the Min8 task at $\text{Start} \leq 5$, $\text{Start} \leq 10$, and $\text{Start} \leq 15$. Station identification, cause classification, and remaining-time estimation are kept separate. Later sections remove the graph, station-group embedding, delay pattern, or split will head, omit process-heuristic labels and clustering, and compare B4 and B5 on the same three start caps.

实验报告 BNPDFormer 在 Min8 任务、 $\text{Start} \leq 5/10/15$ 三档上的结果。工位识别、原因分类和剩余时间估计分开给出。后续各节分别去掉图结构、工位分组嵌入、延迟 pattern 或拆分的 will 头，去掉过程启发式标签和聚类，并在同样三个 Start 档位上与 B4、B5 对照。

1 Data and Protocol / 数据与实验协议

1.1 Task and Evaluation Protocol / 任务与评测协议

The experiments evaluate station-level bottleneck-event prediction at three start horizons, $\text{Start} \leq 5$, $\text{Start} \leq 10$, and $\text{Start} \leq 15$ min, while requiring each target event to last at least eight minutes (Min8). The model observes a 30-min history and evaluates a 20-grid future label range at a one-minute resolution. An *ongoing* target is already active at the last history grid; an *upcoming* target starts after the observation boundary.

实验在三个起始时间范围内评估工位级瓶颈事件预测： $\text{Start} \leq 5$ 、 $\text{Start} \leq 10$ 和 $\text{Start} \leq 15$ 分钟，同时要求每个目标事件至少持续 8 分钟 (Min8)。模型观察 30 分钟历史，并以 1 分钟为时间分辨率，在未来 20 个网格的标签范围内进行评测。若目标在历史序列最后一个网格已经发生，则记为进行中 (ongoing)；若目标在观测边界之后才开始，则记为即将发生 (upcoming)。

The station-identification metrics are will15/who precision, recall, and F1. The name will15 is retained by the implementation but refers to the active Start cap, rather than always denoting a fixed 15-min task. The stricter report metric counts a true positive only when the station is correct and $|\hat{s} - s| \leq 3$ min. Thus, $R_{\text{up}}^{\text{who}}$ measures station detection, whereas $R_{\text{up}}^{\text{rep}}$ additionally enforces start-time tolerance. Remaining-order-time and event-duration errors are reported as mean absolute error (MAE; lower is better). Process-cause performance is reported by accuracy, macro recall, and per-class recall.

工位识别指标包括 will15/who 的精确率、召回率和 F1。代码沿用了 will15 这一名称，但该指标实际对应当前采用的 Start 上限，并不总是表示固定的 15 分钟任务。更严格的 report 指标只有在工位预测正确且 $|\hat{s} - s| \leq 3$ 分钟时才计为真阳性。因此， $R_{\text{up}}^{\text{who}}$ 衡量是否识别出正确工位，而 $R_{\text{up}}^{\text{rep}}$ 还额外要求起始时间满足容差。订单剩余时间和事件持续时间采用平均绝对误差 (MAE，越低越好)；过程原因识别则报告准确率、宏平均召回率和逐类别召回率。

All model and threshold decisions use validation data. The selected checkpoint and threshold are then frozen for test evaluation. Results from different Start caps are not pooled because their positive sets and upcoming denominators differ. All reported runs use seed 42; consequently, no claim of statistical significance is made.

所有模型选择和阈值选择均只使用验证集。确定 checkpoint 和阈值后，将二者冻结并用于测试集评测。由于不同 Start 上限对应的正例集合和 upcoming 分母不同，因此不合并不同 Start 档位的结果。所有实验均只使用 seed 42，所以本文不声称结果具有统计显著性。

1.2 Data Splits and Experiment Families / 数据划分与实验系列

Two main-model settings share the BNPDFormer backbone. Separate report fine-tuning selects each Start cap on its own. Sequential station fine-tuning continues from the previous cap and selects on station F1. The full list of changed settings is in Section 3.

两组主模型设置使用同一个 BNPDFormer 骨干。分档报告微调按 Start 档位分别选模。顺序工位微调沿用上一档 checkpoint，并按工位 F1 选模。具体改动列在第 3 节。

Table 1: Experiment families and protocol boundaries; episode counts are train/validation/test. / 实验系列及协议边界；episode 数依次为训练/验证/测试。

Family	Dataset	Episodes	Main selection metric	Training budget	Test status
Sequential station (retained)	dense_i1	140/28/36	validation will15_f1	100 epochs, patience 40	evaluated
Separate report	dense_i1	140/28/36	validation report_f1	100 epochs, patience 100	evaluated
No heuristic labels or clustering	dense_i1_mix	166/37/45	validation will15_f1	100 epochs, patience 40	evaluated
Structure ablations	dense_i1	140/28/36	validation will15_f1	50 epochs, from scratch	evaluated
B4 BTGCN / B5 BSTAN	fixed 208-episode set	138/30/40	validation will15_f1	100 epochs, patience 40	test unused

2 Implementation Details / 实现细节

The BNPDFormer encoder contains five blocks with an embedding dimension of 80 and an eight-dimensional Laplacian positional encoding. Graph, semantic, and temporal attention use 2, 2, and 4 heads, respectively; these are separate attention modules and should not be described as a single eight-head module. The event head has hidden dimension 80. Training uses batch size 16, a 30-min input history, one-minute windows, and a 20-grid future label range.

BNPDFormer 编码器包含 5 个块，嵌入维度为 80，并使用 8 维拉普拉斯位置编码。图注意力、语义注意力和时间注意力分别使用 2、2 和 4 个头；它们属于不同的注意力模块，因此不能表述为一个统一的“8 头注意力模块”。事件预测头的隐藏维度为 80。训练 batch size 为 16，输入历史长度为 30 分钟，时间窗口为 1 分钟，未来标签范围为 20 个网格。

Optimization uses AdamW with a fine-tuning learning rate of 5×10^{-5} , weight decay 0.05, cosine learning-rate decay, and minimum learning rate 10^{-6} . The final runs allow at most 100 epochs and use early-stopping patience 40. Event-positive windows are oversampled by a factor of 4. The model is initialized from the earlier recall-oriented dense_i1_a1_prefix8 checkpoint, unfreezes the last five encoder blocks and occupancy decoder, and jointly optimizes event, occupancy, remaining-time, and cause objectives. STGNPP and the legacy score/will/mark/tts heads are disabled in the retained configuration.

优化器采用 AdamW，微调学习率为 5×10^{-5} ，权重衰减为 0.05，使用余弦学习率衰减，最低学习率为 10^{-6} 。最终实验最多训练 100 个 epoch，早停 patience 为 40；事件正例窗口按 4 倍过采样。模型从较早的召回导向 checkpoint dense_i1_a1_prefix8 初始化，解冻最后 5 个编码器块和占用解码器，并联合优化事件、占用、剩余时间和原因任务。最终配置关闭 STGNPP，以及旧版 score/will/mark/tts 预测头。

The retained setting is sequential station fine-tuning, configured by FactoryBN_dense_12_3_start{5,10,15}_min8_opt.json. Its runner loads the prefix8 checkpoint at $\text{Start} \leq 5$, then warm-starts $\text{Start} \leq 10$ from the $\text{Start} \leq 5$ checkpoint and $\text{Start} \leq 15$ from $\text{Start} \leq 10$. Checkpoint selection uses validation station F1. The threshold is chosen on validation and then frozen.

保留的设置是顺序工位微调，配置为 FactoryBN_dense_12_3_start{5,10,15}_min8_opt.json。训练脚本在 $\text{Start} \leq 5$ 加载 prefix8 checkpoint，再让 $\text{Start} \leq 10$ 从 $\text{Start} \leq 5$ 热启动， $\text{Start} \leq 15$ 从 $\text{Start} \leq 10$ 热启动。选模使用验证集工位 F1。阈值在验证集上确定后冻结。

3 Main Results / 主实验结果

3.1 Separate Report Fine-tuning and Sequential Station Fine-tuning / 分档报告微调与顺序工位微调

The two settings use the same BNPDFormer, the same dense_i1 split, and the same Min8 task. Separate report fine-tuning is FactoryBN_dense_12_2_start{5,10,15}_min8_ep100.json. Each Start cap loads the prefix8 checkpoint on its own and keeps the parent recipe: checkpoints are chosen by strict report_f1, patience is 100, and an ongoing station is lifted to the report threshold only when its predicted duration is at least eight minutes. Sequential station fine-tuning extends that $\text{Start} \leq 5$ file. run_start_horizon_min8_ep100.py still starts $\text{Start} \leq 5$ from prefix8, then passes the previous best checkpoint to the next cap. Selection changes to station will15_f1.

两组设置使用同一个 BNPDFormer、同一份 dense_i1 划分和同一个 Min8 任务。分档报告微调对应 FactoryBN_dense_12_2_start{5,10,15}_min8_ep100.json。每个 Start 档位各自加载 prefix8 checkpoint，并保留父配置：按严格 report_f1 选模，patience 为 100；进行中工位只有在预测持续时间不少于 8 分钟时，才会被抬到报告阈值。顺序工位微调继承该 $\text{Start} \leq 5$ 配置。run_start_horizon_min8_ep100.py 仍让 $\text{Start} \leq 5$ 从 prefix8 开始，再把上一档最优 checkpoint 传给下一档。选模改为工位 will15_f1。

Both settings already share the validation gate $P \geq 0.80$ and $R \geq 0.70$, split will heads, forced ongoing decisions, and the occupancy/upcoming union. The parent upcoming positive weight is 8 and the far-start positive weight is unset, so it stays 0. Events whose labeled start is within five minutes then receive the shared near-start weight 6, which replaces the upcoming weight. The weights 9/10/11 and the far-start loss boost of 2 therefore reach only events that start later than five minutes. They do not change the $\text{Start} \leq 5$ positive-class weights.

两边都已经带有验证集门槛 $P \geq 0.80$ 、 $R \geq 0.70$ 、拆分的 will 头、进行中强制判定，以及占用与 upcoming 的并集。父配置的 upcoming 正例权重是 8，远起点正例权重没有设置，因此保持为 0。标注起点在 5 分钟以内的事件会改用两边共用的近起点权重 6，并覆盖 upcoming 权重。所以 9/10/11 这组权重和 2 倍远起点损失只作用在起点晚于 5 分钟的事件上，不改变 $\text{Start} \leq 5$ 的正例权重。

Table 2: Training and selection differences. Unlisted fields are inherited unchanged. / 训练与选模差异。未列出的字段沿用父配置，没有改动。

Item	Separate report fine-tuning	Sequential station fine-tuning
Initialization	each cap from prefix8	Start ≤ 5 from prefix8, then $5 \rightarrow 10 \rightarrow 15$
Checkpoint metric	report_f1	will15_f1
Patience	100	40
Upcoming / far positive weight	8 / 0 at all caps	9/9, 10/10, 11/11
Far-start loss boost	1	2, only if start > 5 min
False-positive weight, hard multiplier	2.4, 2.0	2.5, 2.2
Ongoing duration requirement	on	off
Remaining-time loss	$w = 0.35$, uniform	$w = 0.5$, progress floor 0.25, power 1.5
Threshold candidates	0.55–0.98, uneven	0.45–0.90 step 0.05, plus 0.94

Table 2 compares the selected checkpoints. Separate report fine-tuning is stronger at Start ≤ 5 . Sequential station fine-tuning is stronger at Start ≤ 10 and Start ≤ 15 , and its mean validation F1 is 0.845 against 0.832. It is retained because that validation margin holds across the longer caps and the same selection rule is used for B4/B5.

表 2 比较入选 checkpoint。分档报告微调在 Start ≤ 5 更强。顺序工位微调在 Start ≤ 10 和 Start ≤ 15 更强，平均验证 F1 为 0.845，分档报告微调为 0.832。保留顺序工位微调，是因为较长预测范围上的验证优势成立，而且 B4/B5 使用同一条选模规则。

Table 3: Selected checkpoints of the two fine-tuning settings. / 两种微调设置的入选 checkpoint。

Start	Family	Val P	Val R	Val F1	Test P	Test R	Test F1 _{who}	Test F1 _{rep}	R_{up}^{rep} (test)	R_{on}^{who} (test)	Best ep.
≤ 5	Separate report	.865	.872	.869	.832	.808	.820	.819	.542	.923	33
≤ 5	Sequential station	.837	.891	.863	.794	.817	.805	.804	.539	.937	39
≤ 10	Separate report	.805	.836	.820	.750	.763	.757	.751	.550	.923	83
≤ 10	Sequential station	.831	.840	.835	.806	.761	.783	.778	.528	.936	41
≤ 15	Separate report	.814	.803	.808	.772	.716	.743	.741	.489	.925	37
≤ 15	Sequential station	.841	.835	.838	.806	.742	.772	.768	.534	.925	42

The separate-report Start ≤ 5 test station F1 is 0.820, above the sequential setting’s 0.805. At Start ≤ 10 and Start ≤ 15 the sequential setting leads on both validation and test. The retained numbers in the following tables are the sequential station fine-tuning runs.

分档报告微调在 Start ≤ 5 的测试工位 F1 为 0.820，高于顺序工位微调的 0.805。Start ≤ 10 和 Start ≤ 15 上，顺序工位微调在验证集和测试集都更高。后面各表中的保留结果来自顺序工位微调。

Table 4: Separate report fine-tuning station-timing test metrics. / 分档报告微调的工位时间测试指标。

Start	Dur. MAE	Start MAE	F1@1	F1@2	F1@3	R_{up}^{who}	Threshold	$n_{up}/n_{on}/n_{all}$
≤ 5	1.93	.13	.801	.814	.819	.545	.90	336/766/1102
≤ 10	1.77	.24	.730	.749	.751	.560	.88	604/768/1372
≤ 15	1.88	.28	.706	.735	.741	.489	.94	708/770/1478

Table 5: Separate report fine-tuning remaining-time results; MAE is in minutes. / 分档报告微调的剩余时间结果；MAE 单位为分钟。

Start	Val remain MAE	Test remain MAE
≤ 5	12.03	10.64
≤ 10	10.86	9.61
≤ 15	11.77	10.20

Table 6: Separate report fine-tuning cause-classification results. / 分档报告微调的原因分类结果。

Start	Val accuracy	Val macro	Test accuracy	Test macro	Support
≤ 5	.952	.769	.965	.969	579
≤ 10	.956	.774	.965	.970	579
≤ 15	.956	.776	.969	.974	579

The train-derived majority-class accuracy is 0.409. All three separate report runs completed 100 epochs.

依据训练集得到的多数类基线准确率为 0.409。三组分档报告微调均训练满 100 个 epoch。

3.2 Final Main-Model Performance / 最终主模型表现

3.2.1 Station-Level Bottleneck Prediction / 工位级瓶颈预测

Table 7: Sequential station fine-tuning results; validation is the selected epoch and test uses the frozen checkpoint/threshold. / 顺序工位微调结果；验证列来自入选 epoch，测试列使用冻结的 checkpoint 和阈值。

Start	Split	P	R	$F1_{\text{who}}$	$F1_{\text{rep}}$	$R_{\text{up}}^{\text{rep}}$	$R_{\text{on}}^{\text{who}}$	n_{up}	n_{on}	n_{all}	Epoch
≤ 5	Val	.837	.891	.863	.863	.739	.969	—	—	—	39
≤ 5	Test	.794	.817	.805	.804	.539	.937	336	766	1102	39
≤ 10	Val	.831	.840	.835	.835	.711	.960	—	—	—	41
≤ 10	Test	.806	.761	.783	.778	.528	.936	604	768	1372	41
≤ 15	Val	.841	.835	.838	.838	.722	.960	—	—	—	42
≤ 15	Test	.806	.742	.772	.768	.534	.925	708	770	1478	42

The test station-level F1 decreases from 0.805 to 0.772 as the task includes more distant upcoming events. Ongoing recall remains between 0.925 and 0.937, whereas upcoming report recall remains around 0.53. For $\text{Start} \leq 15$, the upcoming bucket recalls are 0.574 for $\text{Start} \leq 5$, 0.522 for starts 6–10, and 0.433 for starts above 10 min. This identifies distant early warning, rather than ongoing detection, as the principal limitation.

随着任务纳入更远的 upcoming 事件，测试集工位级 F1 从 0.805 下降到 0.772。进行中召回率保持在 0.925–0.937，而 upcoming 严格报告召回率约为 0.53。在 $\text{Start} \leq 15$ 设置下， $\text{Start} \leq 5$ 、6–10 分钟以及大于 10 分钟三个 upcoming 桶的召回率分别为 0.574、0.522 和 0.433。这说明主要限制是远距离提前预警能力，而不是进行中事件检测。

3.2.2 Remaining-Time Prediction / 剩余时间预测

Table 8: Sequential station fine-tuning remaining-order-time results; MAE is in minutes. / 顺序工位微调的订单剩余时间结果；MAE 单位为分钟。

Start	Split	Remain MAE
≤ 5	Val	12.3
≤ 5	Test	10.9
≤ 10	Val	12.6
≤ 10	Test	11.2
≤ 15	Val	11.0
≤ 15	Test	9.7

Test remaining-time MAE is 10.9, 11.2, and 9.7 minutes at $\text{Start} \leq 5/10/15$. The error does not grow with the station-prediction horizon: the $\text{Start} \leq 15$ checkpoint is the most accurate of the three.

$\text{Start} \leq 5/10/15$ 的测试集剩余时间 MAE 分别为 10.9、11.2 和 9.7 分钟。该误差并不随工位预测范围变长而上升： $\text{Start} \leq 15$ 的 checkpoint 在三档中最准确。

3.2.3 Cause Classification / 瓶颈原因分类

Table 9: Sequential station fine-tuning cause-classification results. / 顺序工位微调的瓶颈原因分类结果。

Start	Split	Cause accuracy	Cause macro recall	Cause support
≤ 5	Val	.956	.963	—
≤ 5	Test	.965	.967	579
≤ 10	Val	.958	.968	—
≤ 10	Test	.962	.965	579
≤ 15	Val	.967	.978	—
≤ 15	Test	.974	.978	579

Table 10: Sequential station fine-tuning per-class cause recall on test; the final two classes have no test support. / 顺序工位微调的测试集逐类别原因召回率；最后两类在测试集中无样本。

Start	Transport delay	Material shortage	Starved upstream	Queue buildup	Blocked downstream	High utilization
≤ 5	.941	.976	.966	.984	no support	no support
≤ 10	.926	.988	.962	.984	no support	no support
≤ 15	.978	.988	.962	.984	no support	no support

The cause head has four supported classes out of six configured classes. The majority-class baseline is derived from train only and equals 0.409. Test accuracy stays between 0.962 and 0.974, well above that baseline. The unsupported classes are retained in the table to make the support boundary explicit rather than inflating a six-class claim.

原因识别头虽然配置了 6 个类别，但只有 4 个类别拥有有效样本。多数类基线只依据训练集确定，其准确率为 0.409。测试准确率保持在 0.962–0.974，明显高于该基线。表中仍保留两个无测试样本的类别，以明确支持度边界，避免将实际四类有效评测夸大为完整六类评测。

4 Ablation Studies / 消融实验

4.1 Ablation without Process-Heuristic Labels and Clustering / 未加入过程启发式标签与聚类的消融

This ablation removes two components of the full method: the process-heuristic labels and the clustering algorithm. It is trained from `FactoryBN_dense_mix_start{5,10,15}_min8_opt.json` on `dense_i1_mix` (248 episodes, split 166/37/45). Selection uses validation `will15_f1`, and the warm-start order is `Start5→Start10→Start15`. $\text{Start} \leq 10/15$ use the ungated validation best because the precision gate did not open.

该消融去掉完整方法中的两项组成：过程启发式标签和聚类算法。实验使用 `FactoryBN_dense_mix_start{5,10,15}_min8_opt.json`，在 `dense_i1_mix` (248 个 episode，划分 166/37/45) 上训练，以验证集 `will15_f1` 选模，并按 `Start5→Start10→Start15` 顺序热启动。 $\text{Start} \leq 10/15$ 的精确率门槛未开启，因此采用无门槛验证集最优 checkpoint。

4.1.1 Station-Level Bottleneck Prediction / 工位级瓶颈预测

Table 11: Station-level metrics without process-heuristic labels and clustering. / 未加入过程启发式标签与聚类时的工位级指标。

Start	Split	P_{rep}	R_{rep}	$F1_{\text{rep}}$	P_{who}	R_{who}	$F1_{\text{who}}$	$R_{\text{up}}^{\text{rep}}$	$R_{\text{up}}^{\text{who}}$	$R_{\text{on}}^{\text{who}}$	Support $n_{\text{up}}/n_{\text{on}}/n_{\text{all}}$	Best/total ep.
≤ 5	Val	.822	.718	.766	.824	.720	.769	.349	—	.898	—	1/41
≤ 5	Test	.772	.701	.735	.774	.702	.737	.299	.301	.888	562/1216/1778	1/41
≤ 10	Val	.728	.682	.704	.742	.695	.718	.460	—	.872	—	58/98
≤ 10	Test	.684	.613	.646	.697	.625	.659	.324	.344	.855	1000/1221/2221	58/98
≤ 15	Val	.746	.657	.699	.760	.669	.712	.445	—	.868	—	30/70
≤ 15	Test	.697	.586	.637	.704	.592	.643	.320	.325	.847	1168/1222/2390	30/70

Test report F1 falls to .735/.646/.637 at $\text{Start} \leq 5/10/15$, which is .069/.132/.131 below the retained model (.804/.778/.768). Upcoming report recall also falls to .299/.324/.320. Station timing on the same checkpoints is duration MAE 2.54/2.18/2.26 min, start MAE .13/.27/.22 min, and $F1@1/2/3$ equal to .712/.729/.735, .626/.643/.646, and .623/.636/.637.

$\text{Start} \leq 5/10/15$ 的测试 report F1 降至 .735/.646/.637，比保留模型的 .804/.778/.768 分别低 .069/.132/.131。Upcoming 严格召回率也降至 .299/.324/.320。同一 checkpoint 的工位时间指标为：持续时间 MAE 2.54/2.18/2.26 分钟，起点 MAE .13/.27/.22 分钟， $F1@1/2/3$ 分别为 .712/.729/.735、.626/.643/.646 和 .623/.636/.637。

4.1.2 Remaining-Time Prediction / 剩余时间预测

Table 12: Remaining-time results without process-heuristic labels and clustering; MAE is in minutes. / 未加入过程启发式标签与聚类时的剩余时间结果；MAE 单位为分钟。

Start	Val MAE	Test MAE	Test MAE_w	Test primary
≤ 5	12.96	11.85	9.71	9.56
≤ 10	11.58	10.95	9.12	9.77
≤ 15	11.64	10.92	9.03	9.57

MAE_w is the progress-weighted global error and “primary” is the configured middle-progress weighted error. Neither replaces the unweighted MAE. Test MAE is higher than the retained model at $\text{Start} \leq 5$ and $\text{Start} \leq 15$ (11.85 versus 10.9, and 10.92 versus 9.7) and slightly lower at $\text{Start} \leq 10$ (10.95 versus 11.2).

MAE_w 是按生产进度加权的全局误差，“primary”是配置指定的中段进度加权误差；二者都不替代未加权 MAE。测试 MAE 在 $\text{Start} \leq 5$ 和 $\text{Start} \leq 15$ 高于保留模型 (11.85 对 10.9, 10.92 对 9.7)，在 $\text{Start} \leq 10$ 略低 (10.95 对 11.2)。

4.1.3 Cause Classification / 瓶颈原因分类

Table 13: Cause-classification results without process-heuristic labels and clustering. / 未加入过程启发式标签与聚类时的原因分类结果。

Start	Val accuracy	Val macro	Test accuracy	Test macro	Support
≤ 5	.931	.942	.944	.964	964
≤ 10	.928	.937	.924	.940	964
≤ 15	.938	.939	.940	.956	964

Table 14: Per-class cause recall on test without process-heuristic labels and clustering. / 未加入过程启发式标签与聚类时的测试集逐类别原因召回率。

Start	Transport delay	Material shortage	Starved upstream	Queue buildup
≤ 5	.962	.989	.906	1.000
≤ 10	.928	.956	.893	.984
≤ 15	.943	.978	.910	.995

The train-derived majority baseline on this split is 0.495. Test accuracy (.944/.924/.940) remains far above that baseline, but it is below the retained model (.965/.962/.974). Together with the station-level drop, the arm shows that leaving out process-heuristic labels and clustering weakens both event detection and cause classification.

该划分上由训练集得到的多数类基线为 0.495。测试准确率 (.944/.924/.940) 仍远高于该基线，但低于保留模型 (.965/.962/.974)。结合工位级下降，该消融表明：不加入过程启发式标签和聚类，会同时削弱事件检测和原因分类。

4.2 Model-Structure Ablations / 模型结构消融

The nograph arm sets `graph_mode=identity`, removing graph edges. The nogroup arm removes station-group/semantic-group embedding. nopattern disables the delayed pattern injection in the geographic attention keys. nosplit uses one shared event-will head for upcoming and ongoing bottlenecks (`split_will_heads=false`); decode-time forcing of ongoing events is left unchanged. All four arms use the same `dense_i1` split as the final model. $\text{Start} \leq 5$ is trained from scratch for at most 50 epochs with an empty `init_ckpt`; $\text{Start} \leq 10$ and $\text{Start} \leq 15$ warm-start from the previous arm checkpoint. This budget differs from the warm-started 100-epoch full model and is reported as a limitation of causal interpretation.

nograph 将 `graph_mode` 设为 `identity`, 从而移除图边; nogroup 移除工位分组/语义分组嵌入; nopattern 关闭地理注意力键中的延迟 pattern 注入; nosplit 让 upcoming 与 ongoing 共用一个事件 will 头 (`split_will_heads=false`), 解码时对 ongoing 的强制保持不变。四组消融都使用与最终模型相同的 `dense_i1` 划分。Start ≤ 5 以空 `init_ckpt` 从头训练, 上限 50 个 epoch; Start ≤ 10 和 Start ≤ 15 从上一级 checkpoint 热启动。该预算不同于热启动且最多训练 100 个 epoch 的完整模型, 因此这是解释消融因果效果时必须保留的限制。

4.2.1 Station-Level Bottleneck Prediction / 工位级瓶颈预测

Table 15: Structure-ablation station metrics at $\text{Start} \leq 5$; Report F1 requires a correct station and a start error within 3 min. / $\text{Start} \leq 5$ 结构消融工位指标; Report F1 要求工位正确且起点误差不超过 3 分钟。

Arm	Split	P	R	$F1_{\text{who}}$	$F1_{\text{rep}}$	$R_{\text{up}}^{\text{rep}}$	$R_{\text{on}}^{\text{who}}$	n_{up}	n_{on}	n_{all}	Best ep.
Full	Val	.837	.891	.863	.863	.739	.969	—	—	—	39
Full	Test	.794	.817	.805	.804	.539	.937	336	766	1102	39
No graph	Val	.823	.658	.731	.731	.115	.937	—	—	—	19
No graph	Test	.832	.678	.747	.746	.101	.930	336	766	1102	19
No group	Val	.782	.765	.774	.774	.383	.962	—	—	—	28
No group	Test	.766	.756	.761	.753	.307	.941	336	766	1102	28
No pattern	Val	.789	.671	.725	.725	.166	.930	—	—	—	15
No pattern	Test	.799	.698	.748	.745	.158	.935	336	766	1102	15
No split	Val	.784	.670	.724	.723	.149	.937	—	—	—	15
No split	Test	.750	.706	.729	.727	.185	.935	336	766	1102	15

Table 16: Structure-ablation station metrics at $\text{Start} \leq 10$. / $\text{Start} \leq 10$ 结构消融工位指标。

Arm	Split	P	R	$F1_{\text{who}}$	$F1_{\text{rep}}$	$R_{\text{up}}^{\text{rep}}$	$R_{\text{on}}^{\text{who}}$	n_{up}	n_{on}	n_{all}	Best ep.
Full	Val	.831	.840	.835	.835	.711	.960	—	—	—	41
Full	Test	.806	.761	.783	.778	.528	.936	604	768	1372	41
No graph	Val	.807	.486	.606	.606	.049	.894	—	—	—	9
No graph	Test	.821	.539	.651	.649	.065	.909	604	768	1372	9
No group	Val	.663	.790	.721	.721	.636	.934	—	—	—	44
No group	Test	.605	.745	.668	.651	.445	.947	604	768	1372	44
No pattern	Val	.598	.659	.643	.627	.360	.939	—	—	—	28
No pattern	Test	.555	.625	.607	.588	.235	.934	604	768	1372	28
No split	Val	.596	.614	.629	.605	.254	.951	—	—	—	33
No split	Test	.546	.618	.599	.580	.204	.947	604	768	1372	33

Table 17: Structure-ablation station metrics at $\text{Start} \leq 15$. / $\text{Start} \leq 15$ 结构消融工位指标。

Arm	Split	P	R	$F1_{\text{who}}$	$F1_{\text{rep}}$	$R_{\text{up}}^{\text{rep}}$	$R_{\text{on}}^{\text{who}}$	n_{up}	n_{on}	n_{all}	Best ep.
Full	Val	.841	.835	.838	.838	.722	.960	—	—	—	42
Full	Test	.806	.742	.772	.768	.534	.925	708	770	1478	42
No graph	Val	.819	.435	.568	.568	.028	.883	—	—	—	11
No graph	Test	.851	.482	.616	.612	.037	.887	708	770	1478	11
No group	Val	.620	.760	.683	.683	.609	.927	—	—	—	50
No group	Test	.567	.690	.623	.604	.395	.922	708	770	1478	50
No pattern	Val	.631	.769	.705	.693	.600	.956	—	—	—	42
No pattern	Test	.539	.681	.616	.601	.397	.947	708	770	1478	42
No split	Val	.627	.690	.672	.657	.453	.951	—	—	—	47
No split	Test	.539	.641	.607	.586	.309	.951	708	770	1478	47

4.2.2 Remaining-Time Prediction / 剩余时间预测

Table 18: Remaining-time MAE (minutes) for the structure ablations. / 结构消融的剩余时间 MAE (分钟)。

Arm	Start	Val MAE	Test MAE
Full	≤ 5	12.3	10.9
No graph	≤ 5	18.2	17.0
No group	≤ 5	13.2	12.7
No pattern	≤ 5	20.1	18.6
No split	≤ 5	20.8	18.9
Full	≤ 10	12.6	11.2
No graph	≤ 10	24.1	23.3
No group	≤ 10	12.5	11.6
No pattern	≤ 10	14.8	13.4
No split	≤ 10	15.6	14.4
Full	≤ 15	11.0	9.7
No graph	≤ 15	21.4	20.8
No group	≤ 15	11.1	10.6
No pattern	≤ 15	12.5	11.1
No split	≤ 15	15.2	13.8

4.2.3 Cause Classification / 瓶颈原因分类

Table 19: Cause-classification metrics for the structure ablations. / 结构消融的原因分类指标。

Arm	Start	Split	Accuracy	Macro recall	Majority baseline	Support
Full	≤ 5	Val	.956	.963	.409	—
Full	≤ 5	Test	.965	.967	.409	579
No graph	≤ 5	Val	.866	.886	.409	—
No graph	≤ 5	Test	.876	.889	.409	579
No group	≤ 5	Val	.932	.952	.409	—
No group	≤ 5	Test	.943	.955	.409	579
No pattern	≤ 5	Val	.726	.725	.409	—
No pattern	≤ 5	Test	.732	.734	.409	579
No split	≤ 5	Val	.853	.874	.409	—
No split	≤ 5	Test	.853	.874	.409	579
Full	≤ 10	Val	.958	.968	.409	—
Full	≤ 10	Test	.962	.965	.409	579
No graph	≤ 10	Val	.741	.755	.409	—
No graph	≤ 10	Test	.734	.746	.409	579
No group	≤ 10	Val	.928	.944	.409	—
No group	≤ 10	Test	.938	.953	.409	579
No pattern	≤ 10	Val	.807	.823	.409	—
No pattern	≤ 10	Test	.793	.805	.409	579
No split	≤ 10	Val	.888	.904	.409	—
No split	≤ 10	Test	.869	.884	.409	579
Full	≤ 15	Val	.967	.978	.409	—
Full	≤ 15	Test	.974	.978	.409	579
No graph	≤ 15	Val	.796	.790	.409	—
No graph	≤ 15	Test	.794	.787	.409	579
No group	≤ 15	Val	.941	.950	.409	—
No group	≤ 15	Test	.938	.949	.409	579
No pattern	≤ 15	Val	.851	.862	.409	—
No pattern	≤ 15	Test	.843	.850	.409	579
No split	≤ 15	Val	.897	.915	.409	—
No split	≤ 15	Test	.869	.890	.409	579

Table 20: Complete per-class cause recall for the full model and structure ablations on test. / 完整模型与结构消融在测试集上的逐类别原因召回率。

Arm	Start	Transport delay	Material shortage	Starved upstream	Queue buildup	Blocked downstream	High utilization
Full	≤ 5	.941	.976	.966	.984	no support	no support
No graph	≤ 5	.971	1.000	.852	.734	no support	no support
No group	≤ 5	.956	1.000	.911	.952	no support	no support
No pattern	≤ 5	.993	1.000	.797	.145	no support	no support
No split	≤ 5	.934	1.000	.806	.758	no support	no support
Full	≤ 10	.926	.988	.962	.984	no support	no support
No graph	≤ 10	.993	.963	.738	.290	no support	no support
No group	≤ 10	.956	1.000	.890	.968	no support	no support
No pattern	≤ 10	.941	1.000	.802	.476	no support	no support
No split	≤ 10	.831	1.000	.857	.847	no support	no support
Full	≤ 15	.978	.988	.962	.984	no support	no support
No graph	≤ 15	.993	.988	.878	.290	no support	no support
No group	≤ 15	.926	1.000	.911	.960	no support	no support
No pattern	≤ 15	.971	1.000	.857	.573	no support	no support
No split	≤ 15	.875	1.000	.823	.863	no support	no support

Removing graph structure causes the largest and most consistent degradation, especially for upcoming recall, remaining-time MAE, and queue-buildup recall. Removing group embeddings and delayed pattern injection also lowers station F1 at every horizon. The nopattern test report-F1 drops are .059/.190/.167, and its queue-buildup recall is substantially lower. Sharing one will head (nosplit) drops test report F1 by .077/.198/.182. Upcoming report recall falls to .185/.204/.309, while ongoing recall stays above .93. Remaining-time MAE rises to 18.9/14.4/13.8 minutes. Cause accuracy stays high (.853/.869/.869) because the cause head is unchanged; the damage is concentrated in station detection. The nogroup validation precision values (.782/.663/.620) fail the $P \geq .80$ gate, and the gate also never opens for nopattern or nosplit; those rows use the ungated validation best. The nogroup Start ≤ 15 checkpoint is the final epoch (50), so it did not plateau within the assigned budget.

移除图结构造成最大且最一致的性能下降，尤其体现在 upcoming 召回率、剩余时间 MAE 和 queue-buildup 类召回率上。移除工位分组嵌入和延迟 pattern 注入也会降低三个 Start 档位的工位 F1。nopattern 的测试 report-F1 分别下降 .059/.190/.167，且 queue-buildup 类召回率明显降低。共用一个 will 头 (nosplit) 使测试 report F1 下降 .077/.198/.182，upcoming 严格召回率降至 .185/.204/.309，而 ongoing 召回率仍高于 .93。剩余时间 MAE 升至 18.9/14.4/13.8 分钟。原因准确率仍较高 (.853/.869/.869)，因为原因头没有改动，损失集中在工位检测。nogroup 的验证集精确率 .782/.663/.620 未达到 $P \geq .80$ ；nopattern 和 nosplit 的精确率门槛也从未开启，这些行采用无门槛验证集最优值。nogroup 的 Start ≤ 15 checkpoint 位于最后一个 epoch (50)，说明该实验在给定预算内尚未进入平台期。

The planned nosem arm (ablation_nosem_start5_seed42) aborted before epoch 1 because a convolution weight had shape [0,80,1,1]; consequently, it has no valid metrics. This failed arm is reported rather than silently excluded.

计划中的 nosem 实验 (ablation_nosem_start5_seed42) 在第 1 个 epoch 之前即终止，原因是卷积权重形状为 [0,80,1,1]，因此没有有效指标。本文明确报告该失败实验，而不是将其静默删除。

5 Baseline Comparison / 基线对比

5.1 Comparable Setup and Limitations / 对齐设置与局限

B4 is BTGCN with a GCN+GRU128 backbone; B5 is BSTAN with scalar GAT+GRU128. Each baseline retains its own near-history input and parent checkpoint and does not receive BNPDFormer-specific branches. Training follows Start5→Start10→Start15. The six runs use the same Start/Min8 semantics, 20 future grids, threshold candidates, 3-min start tolerance, and validation checkpoint rule as the final BNPDFormer protocol. All six use the fallback F1 rule because none meets both $P \geq .80$ and $R \geq .70$.

B4 为采用 GCN+GRU128 骨干的 BTGCN，B5 为采用标量 GAT+GRU128 骨干的 BSTAN。每个基线保留自身的 near 历史输入和父 checkpoint，不引入 BNPDFormer 专有分支。训练顺序为 Start5→Start10→Start15。六组实验与最终 BNPDFormer 采用相同的 Start/Min8 语义、20 个未来网格、阈值候选集合、3 分钟起点容差以及验证集选模规则。由于六组都未同时满足 $P \geq .80$ 和 $R \geq .70$ ，最终均采用 F1 fallback 规则。

The baseline split is 138/30/40 rather than 140/28/36, and the 40 baseline test episodes have not been evaluated. Therefore, Tables 20 and 21 are complete validation/training reports, but they are not test-set evidence that BNPDFormer outperforms B4/B5.

基线的数据划分为 138/30/40，而不是主模型的 140/28/36，并且基线的 40 个测试 episode 尚未参与评测。因此，表 20 和表 21 是完整的验证集/训练集报告，但不能作为 BNPDFormer 在测试集上优于 B4/B5 的证据。

Table 21: All official B4/B5 validation metrics; hit counts are over upcoming support. / B4/B5 全部正式验证指标; 命中数均以 upcoming 支持数为分母。

Model	Start	P	R	F1	$R_{\text{up}}^{\text{rep}}$	$R_{\text{up}}^{\text{who}}$	$R_{\text{on}}^{\text{who}}$	Upcoming strict	Upcoming station	Best/total ep.	Threshold
B4	≤ 5	.8048	.5655	.6643	.0742	.0742	.8078	32/431	32/431	1/41	.50
B5	≤ 5	.8133	.5709	.6709	.0812	.0812	.8124	35/431	35/431	2/42	.70
B4	≤ 10	.8152	.3829	.5211	.0353	.0353	.6833	27/764	27/764	3/43	.90
B5	≤ 10	.8118	.4163	.5503	.0314	.0340	.7466	24/764	26/764	2/42	.85
B4	≤ 15	.6684	.3718	.4778	.0280	.0325	.7133	25/892	29/892	5/45	.94
B5	≤ 15	.8036	.3543	.4918	.0291	.0336	.6772	26/892	30/892	2/42	.90

Table 22: B4/B5 training station metrics. / B4/B5 训练集工位指标。

Model	Start	Train F1	Train upcoming strict hits	Train R_{up}^{rep}
B4	≤ 5	.6918	98/1753	.0559
B5	≤ 5	.7216	170/1753	.0970
B4	≤ 10	.6349	324/3109	.1042
B5	≤ 10	.6642	295/3109	.0949
B4	≤ 15	.8816	2657/3624	.7332
B5	≤ 15	.6434	477/3624	.1316

Table 23: B4/B5 validation remaining-time MAE; values are middle-progress-weighted and kept in window units. / B4/B5 验证集剩余时间 MAE; 数值为中段进度加权, 单位保留为窗口。

Model	Start	Val remain MAE
B4	≤ 5	13.3421
B5	≤ 5	11.2884
B4	≤ 10	16.2363
B5	≤ 10	10.6643
B4	≤ 15	11.7820
B5	≤ 15	9.6109

Table 24: B4/B5 validation cause-classification metrics. / B4/B5 验证集原因分类指标。

Model	Start	Val cause accuracy	Val cause macro recall
B4	≤ 5	.8876	.8631
B5	≤ 5	.9575	.9538
B4	≤ 10	.9001	.8750
B5	≤ 10	.9572	.9481
B4	≤ 15	.9040	.8837
B5	≤ 15	.9518	.9362

B5 has higher validation F1 than B4 by 0.0066, 0.0292, and 0.0140 at the three horizons and lower remaining-time MAE in every pair. It does not dominate strict upcoming recall: at $\text{Start} \leq 10$, B5 obtains 0.0314 versus B4’s 0.0353. Both baselines remain weak on upcoming events. The most severe generalization gap occurs for B4 $\text{Start} \leq 15$, where train F1/upcoming strict recall are 0.8816/0.7332, but validation F1/upcoming strict recall fall to 0.4778/0.0280.

B5 在三个 Start 档位上的验证 F1 分别比 B4 高 0.0066、0.0292 和 0.0140, 且每一档的剩余时间 MAE 都更低。但 B5 并未在严格 upcoming 召回率上全面占优: $\text{Start} \leq 10$ 时, B5 为 0.0314, 低于 B4 的 0.0353。两个基线在 upcoming 事件上都较弱。最严重的泛化差距出现在 B4 $\text{Start} \leq 15$: 训练 F1/严格 upcoming 召回率为 0.8816/0.7332, 验证集则降至 0.4778/0.0280。

5.2 Overall Discussion and Validity Boundaries / 总体讨论与有效性边界

The retained sequential station fine-tuning reaches test station-level F1 of 0.805/0.783/0.772 and strict upcoming recall of 0.539/0.528/0.534 for $\text{Start} \leq 5/10/15$. The graph, station-group, pattern, and shared-head ablations all reduce station F1. Graph removal nearly eliminates upcoming recall at long horizons, and the shared will head cuts upcoming report recall to 0.185/0.204/0.309. Leaving out process-heuristic labels and clustering lowers test report F1 to 0.735/0.646/0.637. Cause classification and remaining-time estimation are reported separately from station identification; neither is folded into the station F1.

保留的顺序工位微调在 $\text{Start} \leq 5/10/15$ 三档上的测试集工位级 F1 分别为 0.805/0.783/0.772, 严格 upcoming 召回率分别为 0.539/0.528/0.534。移除图结构、工位分组、延迟 pattern, 或让 upcoming 与 ongoing 共用 will 头, 都会降低工位 F1。移除图结构几乎使长预测范围的 upcoming 召回能力消失; 共用 will 头则把 upcoming 严格召回率降到 0.185/0.204/0.309。不加入过程启发式标签和聚类时, 测试 report F1 降至 0.735/0.646/0.637。原因分类和剩余时间估计与工位识别分开报告, 并不并入工位 F1。

Four validity limits must accompany these results. First, all runs use one seed, so uncertainty and significance are unknown. Second, B4/B5 currently have no test result and use a different episode split; their validation scores cannot be presented as a final test ranking. Third, the structure ablations use a smaller, from-scratch budget than the warm-started full model. Fourth, remaining-time metrics are not uniformly comparable: BNPDFormer reports minute-scale `remain_len_mae`, while B4/B5 retain middle-progress-weighted values in window units. These boundaries are preserved in the tables rather than being hidden by a single pooled score.

这些结果必须同时说明四项有效性限制。第一, 所有实验只使用一个随机种子, 因此无法估计不确定性或统计显著性。第二, B4/B5 尚无测试集结果, 且采用不同的 episode 划分, 所以不能将其验证分数作为最终测试排名。第三, 结构消融采用比完整模型更小的、从头训练的预算。第四, 剩余时间指标并非完全可比: BNPDFormer 报告以分钟为单位的 `remain_len_mae`, B4/B5 则保留以窗口为单位的分段进度加权重。本文在各表中明确保留这些边界, 而不是用一个混合总分掩盖差异。